Exercise 6.3

1.

Items	EOQ in units	Minimum Inventory Cost	EOQ in ₹	EOQ in years of supply	Number of order per year
A	2000	₹4	40	2.5	0.4
B	200	₹20	200	0.5	2
C	2627	₹52.54	525.40	0.19	5.26

2 (i) ≈ 913 units per order (ii) ≈ ₹20, 066 per week.

Exercise 6.4

1. $\frac{\partial z}{\partial x} = a(cy + d), \frac{\partial z}{\partial y} = c(ax + b)$

Exercise 6.5

1. 23, 25 3. –2, 8 4. 0.8832, 2.2081 5. $-\frac{4}{3}, -8$ 6. $\frac{10}{79}, -\frac{3}{79}$

Exercise 6.6

1	2	3	4	5	6	7	8	9	10
(d)	(a)	(b)	(a)	(b)	(a)	(b)	(c)	(b)	(a)
11	12	13	14	15	16	17	18	19	20
(a)	(c)	(b)	(b)	(a)	(b)	(b)	(b)	(b)	(c)

Miscellaneous Problems

1. $AC = \frac{10}{x} - 4x^2 + 3x^3, MC = -12x^2 + 12x^3, MAC = \frac{-10}{x^2} - 8x + 9x^2$

2.(i) $\eta_s = \frac{1}{x+1}$ (ii) $\eta_d = \frac{3}{x}$

3. $\eta_s = \frac{4p^2}{2p^2+5}, \frac{4}{7}$

6. TC is increasing in [0.20] ∪ [30, ∞) and decreasing in [20, 30]

7. $x = 3$, TC is minimum

7. FINANCIAL MATHEMATICS

Exercise 7.1

1. ₹ 68,429 2. ₹ 1,20,800 3. ₹ 18,930 4. ₹ 500
5. ₹ 23,79,660 6. ₹ 14,736 7. ₹ 8,433 8. ₹ 1,17,612
9. ₹ 14,339 10. ₹ 1,000

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